ClipSAT : ACT International Subject Test: Mathematics 2 Level 2 [ANSWER KEY]

Unit 4.01: Advanced Functions, Polynomials & Logarithms

Q1. The graph of the function fx = x - 1^2 - 4 is shown in the standard x, y coordinate plane below. What is the range of f ?

[x] A: [-4, infty

[ ] B: [1, infty

[ ] C: -infty, -4]

[ ] D: -infty, infty

Solution: 1. **Identify the vertex and direction of opening**: The function fx = x - 1^2 - 4 is written in vertex form, fx = ax - h^2 + k , where a = 1 , h = 1 , and k = -4 . The vertex of the parabola is at 1, -4. 2. **Determine the minimum output value**: Since the leading coefficient a = 1 > 0 , the parabola opens upward, meaning that the vertex represents the absolute minimum point on the graph. The minimum value attained by fx is -4. 3. **Determine the range**: The range represents all possible output values y-values of the function. Because the function takes on its minimum value of -4 at x = 1 and increases without bound as x moves away from 1, the range is all real numbers greater than or equal to -4, which is written in interval notation as [-4, infty. **Common Misconceptions:** - Choice B, [1, infty, incorrectly uses the x-coordinate of the vertex instead of the y-coordinate. - Choice C, -infty, -4], incorrectly assumes the parabola opens downward. - Choice D, -infty, infty, confuses the range the set of y-values with the domain the set of all real x-values.

Q2. The polynomial function Px = 2x^3 + ax^2 - 5x + b has x - 1 as a factor. When Px is divided by x + 2 , the remainder is -18 . What is the remainder when Px is divided by x - 3 ?

[x] A: 2

[ ] B: 8

[ ] C: 18

[ ] D: -30

Solution: According to the Factor Theorem and the Remainder Theorem, if a polynomial Px is divided by x - c , the remainder is Pc . If x - c is a factor, then Pc = 0 . Step 1: Set up the first equation using the factor x - 1 : P1 = 21^3 + a1^2 - 51 + b = 0 2 + a - 5 + b = 0 implies a + b = 3 Step 2: Set up the second equation using the divisor x + 2 = x - -2 with remainder -18 : P-2 = 2-2^3 + a-2^2 - 5-2 + b = -18 2-8 + 4a + 10 + b = -18 -16 + 10 + 4a + b = -18 implies 4a + b = -12 Step 3: Solve the system of linear equations for a and b : Subtracting the first equation from the second: 4a + b - a + b = -12 - 3 3a = -15 implies a = -5 Substitute a = -5 into a + b = 3 : -5 + b = 3 implies b = 8 Thus, the polynomial is Px = 2x^3 - 5x^2 - 5x + 8 . Step 4: Find the remainder when Px is divided by x - 3 by evaluating P3 : P3 = 23^3 - 53^2 - 53 + 8 = 227 - 59 - 15 + 8 = 54 - 45 - 15 + 8 = 2 Common Pitfalls: - Evaluating P2 instead of P-2 produces 4a + b = -24 , leading to a = -9 , b = 12 , and an incorrect remainder of -30 . - Confusing the constant term b = 8 with the final answer yields 8.

Q3. A function f is defined for all real numbers except x = 0 and x = 1 . For all values of x in its domain, f satisfies the functional equation fx + fleftfrac{x - 1}{x}right = 1 + x. What is the value of f2 ?

[x] A: frac{3}{4}

[ ] B: frac{5}{4}

[ ] C: frac{7}{4}

[ ] D: frac{9}{4}

Solution: To find f2 , substitute strategic values of x to generate a system of equations involving the cyclic transformation gx = frac{x - 1}{x} . 1. Substitute x = 2 : frac{2 - 1}{2} = frac{1}{2} f2 + fleftfrac{1}{2}right = 1 + 2 = 3 quad text{--- Equation 1} 2. Substitute x = frac{1}{2} : frac{frac{1}{2} - 1}{frac{1}{2}} = frac{-frac{1}{2}}{frac{1}{2}} = -1 fleftfrac{1}{2}right + f-1 = 1 + frac{1}{2} = frac{3}{2} quad text{--- Equation 2} 3. Substitute x = -1 : frac{-1 - 1}{-1} = frac{-2}{-1} = 2 f-1 + f2 = 1 + -1 = 0 quad text{--- Equation 3} Now solve this system of 3 linear equations for f2 : From Equation 3, we have f-1 = -f2 . Substituting this into Equation 2 gives: fleftfrac{1}{2}right - f2 = frac{3}{2} quad text{--- Equation 4} Subtract Equation 4 from Equation 1: left[f2 + fleftfrac{1}{2}rightright] - left[fleftfrac{1}{2}right - f2right] = 3 - frac{3}{2} 2f2 = frac{3}{2} implies f2 = frac{3}{4}. Common misconception: Solving for fleftfrac{1}{2}right instead yields frac{9}{4} , which corresponds to a common

Q4. Given the functions fx = 3x - 4 and gx = x^2 + 5 , what is the value of fg2 ?

[x] A: 23

[ ] B: 9

[ ] C: 18

[ ] D: 27

Solution: To find the value of the composite function fg2 , evaluate the inner function first, and then substitute that result into the outer function: 1. Evaluate g2 : g2 = 2^2 + 5 = 4 + 5 = 9 2. Evaluate f9 : f9 = 39 - 4 = 27 - 4 = 23 Common Misconceptions: - Evaluating in the reverse order, gf2 , yields f2 = 32 - 4 = 2 , and g2 = 2^2 + 5 = 9 . - Interpreting the expression as the product f2 cdot g2 yields 2 cdot 9 = 18 . - Evaluating 3 cdot g2 but omitting the constant -4 yields 27 .

Q5. The graph of the quadratic function fx = ax^2 + bx + c is shown in the standard x, y coordinate plane below, with vertex 2, -1 and y-intercept 0, 3. The function g is defined by gx = fx + k - 8 , where k is a constant. If the graph of g passes through the origin 0, 0, which of the following could be the value of k ?

[ ] A: -5

[ ] B: -3

[ ] C: 1

[x] D: 5

Solution: Step 1: Find the equation of fx . Using the vertex form of a parabola with vertex h, k_v = 2, -1: fx = ax - 2^2 - 1 Since the graph passes through the y-intercept 0, 3, substitute x = 0 and y = 3 : 3 = a0 - 2^2 - 1 implies 3 = 4a - 1 implies 4a = 4 implies a = 1 Thus, fx = x - 2^2 - 1 . Step 2: Apply the condition that g passes through 0, 0. Since g0 = 0 and gx = fx + k - 8 : g0 = f0 + k - 8 = fk - 8 = 0 implies fk = 8 Step 3: Solve for k . Substitute k into f : k - 2^2 - 1 = 8 k - 2^2 = 9 k - 2 = 3 quad text{or} quad k - 2 = -3 k = 5 quad text{or} quad k = -1 Among the given choices, 5 is the correct value. Common Algebraic Traps: - Choice A -5 arises from a sign error when writing the vertex form as ax + 2^2 - 1 , yielding k + 2^2 = 9 implies k = -2 - 3 = -5 . - Choice B -3 arises from omitting the horizontal shift of the vertex and setting k = -3 directly from pm 3 . - Choice C 1 arises from solving k + 2 = 3 , or confusing k with an x-intercept of fx .

Q6. What is the minimum value of the function fx = sqrt{x^2 - 4x + 13} + sqrt{x^2 - 14x + 130} for all real numbers x?

[ ] A: 12

[ ] B: 4sqrt{10}

[x] C: 13

[ ] D: sqrt{170}

Solution: To find the minimum value of fx, synthesize the algebraic expressions with coordinate geometry by completing the square under each radical: 1. Rewrite the radicands: x^2 - 4x + 13 = x - 2^2 + 9 = x - 2^2 + 3^2 x^2 - 14x + 130 = x - 7^2 + 81 = x - 7^2 + 9^2 2. Interpret geometrically using the Euclidean distance formula d = sqrt{x_2 - x_1^2 + y_2 - y_1^2}: - sqrt{x - 2^2 + 3^2} is the distance from the point Px, 0 on the x-axis to the fixed point A2, 3. - sqrt{x - 7^2 + 9^2} is the distance from Px, 0 to the fixed point B7, 9. Thus, fx = PA + PB. We are looking for the point Px, 0 that minimizes the sum of distances PA + PB. 3. Apply the reflection principle and the Triangle Inequality: Reflect point A2, 3 across the x-axis to obtain A'2, -3. For any point P on the x-axis, PA = PA'. Therefore: PA + PB = PA' + PB ge A'B with equality occurring if and only if P lies on the straight line segment connecting A'2, -3 and B7, 9. 4. Calculate the length of segment A'B: A'B = sqrt{7 - 2^2 + 9 - -3^2} = sqrt{5^2 + 12^2} = sqrt{25 + 144} = sqrt{169} = 13 The line segment connecting A'2, -3 and B7, 9 intersects the x-axis at x = frac{13}{4} = 3.25, which is a valid real value of x. Hence, the minimum value is exactly 13. **Common Misconceptions:** - **Choice A 12:** Adding the individual minimums of the two radical expressions: minsqrt{x-2^2+9} = 3 at x=2 and minsqrt{x-7^2+81} = 9 at x=7. Because these minima occur at different values of x, they cannot be attained simultaneously. - **Choice B 4sqrt{10}:** Miscalculating the horizontal distance as Delta x = 7 - 3 = 4 rather than 7 - 2 = 5, leading to sqrt{4^2 + 12^2} = sqrt{160} = 4sqrt{10}. - **Choice D sqrt{170}:** Erroneously combining the two radicands directly under a single square root.

Q7. The function f is defined by fx = frac{3x + 2}{x - 4} for all real numbers x neq 4 . What is the value of f^{-1}2 ?

[x] A: -10 | [ ] B: -6
[ ] C: -4 | [ ] D: 6

Solution: By definition of the inverse function, finding k = f^{-1}2 is equivalent to solving fk = 2 . Set up the equation: frac{3k + 2}{k - 4} = 2 Multiply both sides by k - 4 where k neq 4 : 3k + 2 = 2k - 4 3k + 2 = 2k - 8 Subtract 2k from both sides: k + 2 = -8 Subtract 2 from both sides: k = -10 Thus, f^{-1}2 = -10 . Common misconceptions: - Evaluating f2 = frac{32 + 2}{2 - 4} = frac{8}{-2} = -4 confuses the function f with its inverse f^{-1} . - A sign error when solving 3k + 2 = 2k - 8 can lead to k = -6 or k = 6 .

Q8. The polynomial function Px = x^3 + ax^2 + bx + 12 , where a and b are real constants, has x - 1 as a factor. When Px is divided by x - 2 , the remainder is -10 . What is the greatest root of the equation Px = 0 ?

[x] A: 4 | [ ] B: 3
[ ] C: 1 | [ ] D: 6

Solution: To find the roots of Px = 0 , we first determine the unknown parameters a and b using the Factor Theorem and the Remainder Theorem. 1. **Apply the Factor Theorem:** Since x - 1 is a factor of Px , we have P1 = 0 : P1 = 1^3 + a1^2 + b1 + 12 = 0 implies 1 + a + b + 12 = 0 implies a + b = -13 2. **Apply the Remainder Theorem:** When Px is divided by x - 2 , the remainder is -10 , so P2 = -10 : P2 = 2^3 + a2^2 + b2 + 12 = -10 implies 8 + 4a + 2b + 12 = -10 implies 4a + 2b = -30 Dividing by 2 gives: 2a + b = -15 3. **Solve the linear system:** Subtract the first equation from the second: 2a + b - a + b = -15 - -13 implies a = -2 Substitute a = -2 back into a + b = -13 : -2 + b = -13 implies b = -11 Thus, the polynomial is: Px = x^3 - 2x^2 - 11x + 12 4. **Factor Px and find the roots:** Since x - 1 is a factor, divide Px by x - 1 : x^3 - 2x^2 - 11x + 12 = x - 1x^2 - x - 12 Factoring the quadratic factor gives: x^2 - x - 12 = x - 4x + 3 So the full factorization is: Px = x - 1x - 4x + 3 The roots of Px = 0 are x = 1 , x = 4 , and x = -3 . The greatest root is 4 . *Common Traps:* - Factoring x^2 - x - 12 incorrectly as x + 4x - 3 leads to roots -4, 1, 3 , with 3 mistakenly identified as the greatest root. - Only noting the given root from the factor x - 1 gives 1 .